SUBGROUP SCHEME CRITERION: FACTORIZATION OF E, M, I THROUGH H

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file groupoids.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: this chapter's pool has 47 of 128 labels already delivered by the S7 line (S7-gp-* registrations, statement_anchor-prose scan; withdrawn cards excluded); none of the 11 accepted labels of this batch is among them (gp-occupied.json, 2026-10-14).

Lemma 0.1. *Let S be a scheme. Let (G, m, e, i) be a group scheme over S.*

(1) *A closed subscheme $H \subset G$ is a closed subgroup scheme if and only if $e : S \to G$, $m|_{H \times_S H} : H \times_S H \to G$, and $i|_H : H \to G$ factor through H.*
(2) *An open subscheme $H \subset G$ is an open subgroup scheme if and only if $e : S \to G$, $m|_{H \times_S H} : H \times_S H \to G$, and $i|_H : H \to G$ factor through H.*

Proof. Looking at T-valued points this translates into the well known conditions characterizing subsets of groups as subgroups. □

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